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SPIN-OUT 360-DEGREE CAMERA FOR SMARTPHONE

Abstract

Camera modules comprising two sub-cameras included in a camera housing and a spin-out actuator. The two sub-cameras are a first sub-camera with first field of view (FOV) FOV1≥180 deg oriented along a camera module optical axis and pointing in a first direction and a second sub-camera having a second FOV, FOV2≥180 deg, the second sub-camera being oriented along the camera module optical axis and pointing in a second direction which is opposite to the first direction. Such a camera module is operational to capture a 360 degree panoramic image or video stream by combining images obtained with the first sub-camera and the second sub-camera, where the spin-out actuator is operational to rotate the camera housing around an axis perpendicular to the camera module optical axis for switching the camera between a stowed position and a spun-out/operational position.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of U.S. patent application Ser. No. 18/194,637 filed Apr. 2, 2023 (now allowed), which claims the benefit of priority from U.S. provisional patent application No. 63/329,384 filed Apr. 9, 2022, which is incorporated herein by reference in its entirety.

FIELD
[0002] The subject matter disclosed herein relates in general to cameras embedded in mobile devices, in particular cameras embedded in smartphones.
BACKGROUND
[0003] Mobile electronic handheld devices ("mobile devices") such as smartphones or tablets having multiple cameras (or "multi-cameras") with different fields of view (FOVs) are ubiquitous. In the following, we use "mobile device" and "smartphone" interchangeably.
[0004] A well-known camera type in the photography field is the omnidirectional camera ("omni" meaning "all"), also known as "360-degree camera" or 360° camera. This is a camera having a FOV that covers an entire 360° sphere. Omnidirectional cameras are important and prevalent in areas where large visual field coverage is needed, such as in panoramic photography as well as in action and sport videography. For example, a 360° camera can be realized by using two sub-cameras, where (i) each of the sub-cameras covers a FOV of 180 degrees or more ("180-degree sub-camera") and (ii) the FOVs of the sub-cameras cover different segments of a scene. By combining (or "stitching") simultaneously captured images from the two sub-cameras, the entire 360° sphere is covered.
[0005] In recent years, there have been attempts to embed a 360° camera in mobile devices. However, the physical (or size) constraints of such cameras have hindered those attempts. In particular, the 180-degree sub-cameras must protrude from the mobile device's housing to be able to cover a FOV of 180 degrees or more. In general, a mobile device's industrial design is optimized for a low thickness and plane surfaces. The need for protrusion of a 360° camera conflicts with the aforementioned requirement for low thickness and plane surfaces and represents a technical challenge.
[0006] It would be advantageous to have a spin-out 360° camera integrated in a mobile device that is switchable between a "stowed" state—in which the camera module is inactive, and a "spun-out" state—in which the camera module is active and operational to capture the entire 360° sphere. The spin-out 360° camera protrudes only when the camera is in use (spun-out) and does not protrude when the camera is not in use (stowed). It is observed that the slimness and flatness are required only when the camera is inactive, e.g. when a mobile device is in a pocket or similar enclosed space. Thus, making the camera module spin-out and stow on request bridges the conflicting requirements.
SUMMARY
[0007] In various exemplary embodiments, there are provided camera modules, comprising: a first sub-camera comprising a first lens having a first effective focal length (EFL.sub.1) and a first image sensor, the first sub-camera having a first field of view FOV1≥180 degrees and being oriented along a camera module optical axis and pointing in a first direction; a second sub-camera comprising a second lens having a second effective focal length (EFL.sub.2)=EFL.sub.1 and a

second image sensor, the second sub-camera having a second field of view $FOV2 \geq 180$ degrees and being oriented along the camera module optical axis and pointing in a second direction which is opposite to the first direction; and a spin-out actuator, wherein the first sub-camera and the second sub-camera are included in a camera housing, wherein the camera is operational to capture a 360 degree panoramic image or video stream by combining images obtained with the first sub-camera and with the second sub-camera, wherein the spin-out actuator is operational to rotate the camera housing around an axis perpendicular to the camera module optical axis for switching the camera between a stowed position and a spun-out position, wherein the camera module is active in the spun-out position.

[0008] In some examples, the camera module is included in a mobile device, wherein in the spun-out operational position the camera module optical axis is perpendicular to a front surface of the mobile device.

[0009] In some examples, the camera module is included in a mobile device, wherein in the stowed position the camera module optical axis is parallel to a front surface of the mobile device.

[0010] In some examples, the camera module is included in a mobile device, wherein in the stowed position the camera module housing is flush with both a front surface and a rear surface of the mobile device.

[0011] In some examples, the camera housing is rotated by 90 degrees for switching between the stowed position and the spun-out position.

[0012] In some examples, the camera module is included in a mobile device, the first image sensor and the second image sensor are mounted on a single printed circuit board.

[0013] In some examples, the camera module has a camera module height HM in the range of 5 mm-20 mm. In some examples, HM is in the range of 7 mm-11 mm.

[0014] In some examples, the camera module has a camera module width WM in the range of 10 mm-30 mm. In some examples, WM is in the range of 15 mm-20 mm.

[0015] In some examples, the spin-out actuator is an actuator selected from the group consisting of an electric stepper motor, a voice coil motor, and a shaped memory alloy actuator.

[0016] In some examples, the camera module comprises a spin-out mechanism to rotate the camera housing, wherein the spin-out mechanism includes a worm-screw and a worm wheel, and wherein the worm-screw engages with the worm wheel.

[0017] In some examples, EFL.sub.1 and EFL.sub.2 are in the range of 0.75 mm-2.5 mm. In some examples, EFL.sub.1 and EFL.sub.2 are in the range of 0.9 mm-1.5 mm. In some examples, EFL.sub.1 and EFL.sub.2 are in the range of 0.75 mm-2.5 mm EFL.sub.1 and EFL.sub.2 are in the range of 1 mm-1.2 mm.

[0018] In some examples, the first lens and the second lens each include N=6 lens elements. In some examples, a power sequence of the N=6 lens elements is negative-negative-positive-positive-negative-positive.

[0019] In some examples, a first lens element L.sub.1 of both the first lens and the second lens is made of glass. In some examples, a last lens element L.sub.6 of both the first lens and the second lens is the strongest lens element in the lens. In some examples, a last lens element L.sub.6 of both the first lens and the second lens is the strongest lens element in the lens.

[0020] In some examples, a f number f.sub.3 of a third lens element L.sub.3 of both the first lens and the second lens fulfills $f.sub.3 < 2 \times EFL$. In some examples, both the first lens and the second lens have a f number lower than 3.

[0021] In some examples, both FOV1 and FOV2 are smaller than 200 degrees.

[0022] In some examples, a camera module is included in a mobile device such as a smartphone.

[0023] In some examples, a camera module is included in a multi-camera. In some examples, the multi-camera is included in a mobile device such as a smartphone.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] Non-limiting examples are described below with reference to figures attached hereto that are listed following this paragraph. Identical structures, elements or parts that appear in more than one figure are generally labelled with a same numeral in all the figures in which they appear. The drawings and descriptions are meant to illuminate and clarify examples of the subject matter disclosed herein, and should not be considered limiting in any way. In the drawings:

[0025] FIG. **1**A illustrates a smartphone including a 360° camera disclosed herein in a stowed state in a front view;

[0026] FIG. **1**B illustrates the smartphone and 360° camera of FIG. **1**A in a spun-out state in a front view;

[0027] FIG. **2**A illustrates the smartphone and 360° camera of FIGS. **1**A-B in a stowed state in one side view;

[0028] FIG. **2**B illustrates the smartphone and 360° camera of FIGS. **1**A-B in a spun-out state in one side view;

[0029] FIG. **3**A illustrates the smartphone and 360° camera of FIGS. **1**A-B in a stowed state in another side view;

[0030] FIG. **3**B illustrates the smartphone and 360° camera of FIGS. **1**A-B in a spun-out state in another side view;

[0031] FIG. **4** illustrates another embodiment of a 360° camera disclosed herein;

[0032] FIG. **5** shows an example of an optical lens system for use in a 360° camera disclosed herein.

DETAILED DESCRIPTION

[0033] FIG. **1**A illustrates a smartphone **100** including a 360-degree camera **110** as disclosed herein in a “stowed” state in a front view. Hereinafter, 360° cameras described below may drop the “360°” term for simplicity. In the stowed state, camera **110** is inactive. Smartphone **100** includes a first area **102** which does not include a screen (“bezel area”), a second area **104** which includes a screen (“screen area”), and a housing (body) **106**.

[0034] A front surface (or “user surface”) of smartphone **100** is visible. “Front surface” of a smartphone is defined here as the surface of the smartphone that includes a screen. Accordingly, a “rear surface” (or “world facing surface”) of a smartphone is defined here as a surface parallel to the front surface, but having a surface normal that points in an opposite direction than a surface normal of the front surface. In general, the rear surface does not include a screen. Camera **110** is included (i.e. embedded) in bezel area **102**. Camera **110** includes two, first and second sub-cameras (not shown here), see **120** and **130** in FIGS. **2**A-B. Optical axes of the first sub-camera and of the second sub-camera of camera **110** (not shown) are oriented parallel to the x-axis in the x-y-z coordinate system shown. In the stowed state, camera **110** is positioned so that it forms a plane surface (i.e. “is flush”) with the surrounding bezel area **102**. In the stowed state, an aperture of the first sub-camera and an aperture of the second sub-camera are not exposed and are therefore protected from damage in case of a drop or an impact.

[0035] FIG. **1**B illustrates smartphone **100** including camera **110** in a “spun-out” state in the same view shown in FIG. **1**A. In the spun-out state, camera **110** is active. A first aperture **116** of camera **110** is visible. The optical axes of the first sub-camera and of the second sub-camera of camera **110** are now oriented parallel to the z-axis. With respect to the stowed state in FIG. **1**A, camera **110** is rotated by 90 degrees around the y-axis. In this (active) spun-out state, camera **110** has an unobstructed FOV of more than 180 degrees, i.e. the FOV of lenses of camera **110** is not obstructed by housing **106** or any other component included in smartphone **100**. In other embodiments for switching a camera such as camera **110** from a stowed state to a spun-out state, the camera may be rotated by 90 degrees around the x-axis.

[0036] In some examples, camera **110** may be included in a multi-camera as known in the art. In a multi-camera, two or more cameras are included that have lenses with different focal lengths to capture images of a same scene with FOVs. For example, in addition to camera **110**, a multi-camera may include a Wide camera having a Wide camera FOV ("FOV.sub.W") of e.g. 80 degrees and a Tele (or "zoom") camera having a narrower FOV ("native FOV.sub.T" or ("n-FOV.sub.T")) of e.g. 25 degrees and with higher spatial resolution (for example 3-5 times higher) than that of the Wide camera. Smartphone **100** may in addition include an application processor ("AP"), e.g. configured to switch between different cameras in the multi-camera, to process image data of the multi-camera, to supply control signals for switching camera **110** from a stowed state to a spun-out state and vice versa etc.

[0037] FIG. **2**A illustrates smartphone **100** including 360° camera **110** of FIGS. **1**A-B in a stowed state in a cross-sectional view. Smartphone **100** has a smartphone module height HM and a smartphone module width WM. Camera **110** includes a housing **112** and a pivot (or "rotation") axis **114**. For switching camera **110** between a stowed (inactive) and a spun-out (operational) state, camera **110** is rotated by 90 degrees around pivot axis **114**. Pivot axis **114** is substantially parallel to the y-axis. A first aperture **116** of first sub-camera **120** and a second aperture **118** of second sub-camera **130** of camera **110** are visible. In the inactive state, both first aperture **116** and second aperture **118** are directed towards the inwards of smartphone **100**, i.e. camera **110** cannot capture a scene. Camera **110** has a camera width of W.sub.C (measured along the y-axis) and a camera height H.sub.C (measured along the z-axis). In some examples, WM may be 10 mm-30 mm and HM may be 5 mm-20 mm. In some examples, WM may be 15-20 mm and HM may be 7-11 mm. In some examples, WM may be 18 mm and HM may be 9 mm. A circle **202** is defined as the smallest circle that fully encircles (or contains) camera **110**.

[0038] FIG. **2**B illustrates smartphone **100** including camera **110** in a spun-out state in a cross-sectional view. Both first aperture **116** and second aperture **118** are directed towards a scene, and both first sub-camera **120** and second sub-camera **130** are operational to capture the scene. A first FOV **122** of first sub-camera **120** and a second FOV **132** of second sub-camera **130** may have identical sizes (values), or may be different. First FOV **122** and second FOV **132** may each cover about 150-210 degrees, or preferably they may cover 180-210 degrees.

[0039] FIG. **3**A and FIG. **3**B illustrate smartphone **100** including camera **110** in another cross-sectional view in, respectively, stowed and spun-out states. Camera **110** includes a spin-out mechanism **302** operational to rotate camera **110** around pivot axis **114** for switching camera **110** between the stowed and spun-out states. Spin-out mechanism **302** includes an actuator **304** that is connected to a worm screw **306**. Worm screw **306** engages with a worm wheel **308** to transmit the actuation of actuator **304** so that camera **110** is rotated around the y-axis at pivot **114**. Actuator **304** may for example be an electric stepper motor, a voice coil motor (VCM), or a shaped memory alloy (SMA) actuator. In the spun-out state, a free space (or volume) **312** is generated.

[0040] FIG. **4** illustrates another example of a 360-degree camera disclosed herein and numbered **400**. Camera **400** includes a first sub-camera **420**, a second sub-camera **430**, and a camera module housing **402**. First sub-camera **420** and second sub-camera **430** include respectively lenses **422** and **432** included in lens barrels **424** and **434** and having lens optical axes **426** and **436**, image sensors **428** and **438**, and apertures **429** and **439**. Image sensor **428** and image sensor **438** are oriented parallel to each other and perpendicular to optical axes **426** and **436**, wherein the active regions (i.e. the regions including imaging pixels) of image sensor **428** and image sensor **438** face opposite directions. Image sensor **428** points towards (or faces) a negative z-direction, and image sensor **438** points towards (or faces) a positive z-direction. A pivot (rotation) axis **414** is substantially parallel to the y-axis. Image sensor **428** and image sensor **438** may both be mounted on a single printed circuit board (PCB) **416**. In other examples, two or more PCBs may be used to mount two image sensors such as image sensor **428** and image sensor **438**. To supply control signals and retrieve image data from camera **400**, PCB **416** or the two or more PCBs may be electrically connected to a smartphone including camera **400** by a flexible electric cable (not shown). A worm wheel **408** may transmit the actuation required for switching camera **400** between a stowed state and a spun-out state.

[0041] FIG. **5** shows an example of an optical lens system disclosed herein and numbered **500**. Optical lens system **500** comprises a lens **502**, an image sensor **504** and, optionally, an optical element ("window") **506**. Optical element **506** may be for example infra-red (IR) filter, and/or a glass image sensor dust cover. Optical lens system **500** may be used in a sub-camera of a 360-degree camera such as first sub-camera **420** and second sub-camera **430**.

[0042] Optical rays pass through lens **502** and form an image on image sensor **504**.

[0043] Detailed optical data and surface data for lens **502** are given in Tables 1-2. Optical lens system **500** has a FOV of 191 degrees, an EFL of 1.08 mm and a f number of 2.72.

[0044] Table 1 provides surface types and Table 2 provides aspheric coefficients. [0045] The surface types are: [0046] a) Plano: flat surfaces, no curvature [0047] b) Q type 1 (QT1) surface sag formula:

[00001] $z(r) = \frac{cr^2}{1+\sqrt{1-(1+k)c^2r^2}} + D_{con}(u) \quad (Eq.1) \; D_{con}(u) = u^4$ .Math.$_{n=0}^{N} A_n Q_n^{con}(u^2)$

$u = \frac{r}{r_{norm}}, x = u^2 \; Q_0^{con}(x) = 1 Q_1^{con} = -(5-6x) Q_2^{con} = 15 - 14x(3-2x)$

$Q_3^{con} = -\{35 - 12x[14 - x(21-10x)]\} Q_4^{con} = 70 - 3x\{168 - 5x[84 - 11x(8-3x)]\}$

$Q_5^{con} = -[126 - x(1260 - 11x\{420 - x[720 - 13x(45-14x)]\})]$ [0048] c) Even Asphere (ASP) surface sag formula:

[00002]

$$z(r) = \frac{cr^2}{1+\sqrt{1-(1+k)c^2r^2}} + {}_1r^2 + {}_2r^4 + {}_3r^6 + {}_4r^8 + {}_5r^{10} + {}_6r^{12} + {}_7r^{14} + {}_8r^{16} \quad (Eq.2)$$

where {z, r} are the standard cylindrical polar coordinates, c is the paraxial curvature of the surface, k is the conic parameter, r.sub.norm is generally one half of the surface's clear aperture ("CA", or also "DA" for clear aperture diameter), and A.sub.n are the aspheric coefficients shown in lens data tables.

[0049] The Z axis is positive towards image. Values for CA are given as a clear aperture radius, i.e. D/2. The reference wavelength is 555.0 nm. Units in Table 1 are in mm except for refraction index ("Index") and Abbe #. Each lens element Li has a respective focal length f.sub.i, both given in Table 1.

TABLE-US-00001 TABLE 1 Optical lens system 500 EFL = 1.08 mm, F# = 2.72, FOV = 191 deg. Curvature Aperture Focal Surface # Comment Type Radius Thickness Radius (D/2) Material Index Abbe # Length 1 Lens 1 Standard 6.529 1.784 4.068 Glass 1.95 32.3 −4.679 2 2.303 0.222 1.725 3 Lens 2 QT1 0.824 0.763 1.358 Plastic 1.58 28.4 −5.858 4 0.436 0.783 0.697 5 A.S. Plano Infinity 0.071 0.219 6 Lens 3 QT1 −43.758 0.759 0.333 Plastic 1.53 55.7 1.607 7 −0.8509 0.058 0.675 8 Lens 4 QT1 12.013 1.052 0.986 Plastic 1.53 55.7 2.741 9 −1.624 0.041 1.024 10 Lens 5 QT1 −1.956 0.294 1.009 Plastic 1.67 19.2 −2.384 11 9.781 0.033 1.147 12 Lens 6 QT1 3.686 0.969 1.495 Plastic 1.54 55.9 1.345 13 −0.832 0.232 1.617 14 Filter Plano Infinity 0.210 — Glass 1.52 64.2 15 Infinity 0.350 — 16 Image Plano Infinity — —

TABLE-US-00002 TABLE 2 Aspheric Coefficients Surface # Norm Radius Conic A0 A1 A2 A3 3 1.352 −2.324 −0.04791 −0.05523 0.015938 −0.00196 4 0.679 −0.645 −0.318956 −0.019675 −0.010019 −0.001377429 5 (SA) 0 0 0 0 0 0 6 0.544 −1.662 0.022799 0.028698 0.010205 −1.46082E−05 7 0.813 −0.098 0.07629 −0.007024 0.00877 −0.007080169 8 1.304 124.913 0.176348 −0.125627 −0.006741 −0.009985724 9 1.140 0.232 −0.217642 0.109874 −0.000286 0.001776936 10 1.431 −0.240 −0.193425 0.330347 −0.11511 −0.022503981 11 1.337 65.327 −0.075806 0.035603 −0.014417 0.005430028 12 1.362 3.497 0.304975 −0.17175 0.057573 −0.02251768 13 2.045 −2.236 1.275921 −0.720937 0.221204 −0.21164209 Aspheric Coefficients (Continued) Surface # A4 A5 A6 A7 A8 3 3.92449E−05 −0.00028 −3.4466E−05 7.11888E−05

−0.00011 4 −0.001139 −0.00034385 −0.00011 −5.09204E−06 3.11037E−06 5 (SA) 0 0 0 0 0 6 −0.007396 −0.0105359 −0.007271 −0.002404784 −9.43331E−05 7 −0.001707 0.000908346 0.002366 −0.000553186 −0.00223188 8 0.01722 0.00132032 −0.002979 −0.006638898 −0.002825671 9 −0.005606 −0.00098302 −0.001488 0.001349993 0.002046746 10 −0.019598 0.023306945 −0.006583 0.012424048 −0.009745867 11 −0.029477 0.012113887 −0.008651 0.007679978 0.003367427 12 0.006928 −0.00083061 −0.00035 0.00076921 −0.000764853 13 −0.03595 −0.05541839 0.088623 0.08797425 0.064227559

[0050] Optical lens system **500** has an effective focal length ("EFL") of 1.08 mm. A power sequence of lenses L.sub.1-L.sub.6 included in lens **502** is negative-negative-positive-positive-negative-positive. L.sub.1 is made of glass. A focal length of lens element L.sub.i is f.sub.i, i=1-6. f.sub.3<2.Math.EFL and f.sub.6<2.Math.EFL or even f.sub.6<1.5.Math.EFL. L.sub.2 is the lens element having a largest focal length magnitude, i.e. |f.sub.2|>|f.sub.i|, i=1, 3, ... In other embodiments, EFL may be in the range of 0.5 mm-5 mm, or even in the range 0.75 mm-2.5 mm, for example 0.9 mm-1.5 mm or 1 mm-1.2 mm.

[0051] For the sake of clarity, the term "substantially" is used herein to imply the possibility of variations in values within an acceptable range. According to one example, the term "substantially" used herein should be interpreted to imply possible variation of 0-10% over or under any specified value.

[0052] It is to be noted that the various features described in the various embodiments can be combined according to all possible technical combinations.

[0053] It is to be understood that the disclosure is not limited in its application to the details set forth in the description contained herein or illustrated in the drawings. The disclosure is capable of other embodiments and of being practiced and carried out in various ways. Hence, it is to be understood that the phraseology and terminology employed herein are for the purpose of description and should not be regarded as limiting. As such, those skilled in the art will appreciate that the conception upon which this disclosure is based can readily be utilized as a basis for designing other structures, methods, and systems for carrying out the several purposes of the presently disclosed subject matter.

[0054] Those skilled in the art will readily appreciate that various modifications and changes can be applied to the embodiments of the disclosure as hereinbefore described without departing from its scope, defined in and by the appended claims.

Claims

1. A mobile device, comprising: a screen; and a camera module, comprising: a first sub-camera having a first lens, a first effective focal length (EFL.sub.1), a first image sensor and a first field of view (FOV1)≥180 degrees, the first sub-camera pointing in a first direction, a second sub-camera having a second lens, a second effective focal length (EFL.sub.2)=EFL.sub.1, a second image sensor and a second field of view (FOV2)≥180 degrees, the second sub-camera pointing in a second direction which is opposite to the first direction, and a spin-out actuator, wherein the first sub-camera and the second sub-camera are included in a camera housing, wherein the camera module is operational to capture a 360 degree panoramic image or video stream by combining images obtained with the first sub-camera and with the second sub-camera, wherein the spin-out actuator is operational to rotate the camera housing for switching the camera between a stowed position and a spun-out position, wherein the camera module is active in the spun-out position, wherein in the spun-out position the first and second directions are perpendicular to the screen.
2. The mobile device of claim 1, wherein EFL.sub.1 and EFL.sub.2 are in the range of 0.75 mm-2.5 mm.
3. The mobile device of claim 1, wherein in the stowed position the first and second directions are parallel to the screen.

4. The mobile device of claim 1, wherein in the stowed position the camera housing is flush with both a front surface and a rear surface of the mobile device.
5. The mobile device of claim 1, wherein the camera module has a camera module height HM in the range of 5 mm-20 mm.
6. The mobile device of claim 5, wherein HM is in the range of 7 mm-11 mm.
7. The mobile device of claim 1, wherein the camera module has a camera module width WM in the range of 10 mm-30 mm.
8. The mobile device of claim 7, wherein WM is in the range of 15 mm-20 mm.
9. The mobile device of claim 1, wherein the spin-out actuator is an actuator selected from the group consisting of an electric stepper motor, a voice coil motor, and a shaped memory alloy actuator.
10. The mobile device of claim 1, wherein the camera module comprises a spin-out mechanism to rotate the camera housing, wherein the spin-out mechanism includes a worm-screw and a worm wheel, and wherein the worm-screw engages with the worm wheel.
11. The mobile device of claim 1, wherein EFL.sub.1 and EFL.sub.2 are in the range of 0.9 mm-1.5 mm.
12. The mobile device of claim 1, wherein EFL.sub.1 and EFL.sub.2 are in the range of 1 mm-1.2 mm.
13. The mobile device of claim 1, wherein each of the first lens and the second lens include N=6 lens elements.
14. The mobile device of claim 13, wherein a power sequence of the N=6 lens elements is negative-negative-positive-positive-negative-positive.
15. The mobile device of claim 1, wherein a first lens element L.sub.i of both the first lens and the second lens is made of glass.
16. The mobile device of claim 1, wherein a last lens element L.sub.6 of both the first lens and the second lens is the strongest lens element in the lens.
17. The mobile device of claim 1, wherein an f number f.sub.3 of a third lens element L.sub.3 of both the first lens and the second lens fulfills $f_3<2\times EFL$.
18. The mobile device of claim 1, wherein both the first lens and the second lens have a f number lower than 3.
19. The mobile device of claim 1, wherein both FOV1 and FOV2 are smaller than 200 degrees.
20. The mobile device of claim 1, wherein the mobile device is a smartphone.